

Secure Face Registration and Verification Framework

Final Project Report

Project Codename: Nishika Report Date: August 6, 2026

1. Problem Statement

This system acts as a security guard for face-based login, with two jobs. First, when someone signs up (registration), it takes a clear photo of their face and remembers it. Second, when someone logs in later (verification), it looks at their face again and answers two questions: is this a real, live person right now, and does this face belong to the person who signed up.

A face-matching system alone can be fooled by a printed photo, a video, or a phone screen. This system adds two more capabilities: liveness detection (is this a real, live person) and image quality assessment (is the photo even usable). The two phases -- registration and verification -- share one pipeline with two entry points, reusing the same quality, liveness, and matching logic.

2. Approach and Architecture

Shared pipeline: Capture (WebRTC webcam frame) -> Face Detection (MediaPipe, rejects 0 or more than 1 face) -> Quality Assessment (7-check weighted composite score) -> Liveness Detection (Passive: MiniFASNet; Active: blink or head-turn challenge) -> Face Embedding (ArcFace via DeepFace, pretrained).

After embedding, the pipeline forks:

- Registration: Consent gate -> Duplicate check (cosine similarity vs. existing templates, threshold 0.68) -> Store template (Fernet AES-128 encrypted at rest, front-facing only).
- Verification: Fetch all active users' templates -> 1-to-N cosine-similarity match, accept at score 0.40 or higher -> Accept/Reject, logged to verification_logs.

Deployment: three local components on one machine, no cloud dependency -- Streamlit UI (with streamlit-webrtc for the camera) -> FastAPI backend (/register, /verify) -> SQLite (local file: users, templates, access_log, verification_logs). API-key auth and a sliding-window rate limiter sit in front of the backend.

Deliberate change from the original brief: registration originally captured three angles (front/left/right); this was simplified to a single front-facing capture -- full rationale in Section 4.

3. Results - Real Measured Numbers

3.1 Face Matching Accuracy (Frontal-vs-Frontal, the production path)

Equal Error Rate (EER): 3.19% at threshold 0.2642
ROC AUC: 0.9953
At threshold 0.40 - False Accept Rate: 0.34% (2/595)
At threshold 0.40 - False Reject Rate: 15.09% (16/106)
Half Total Error Rate (HTER): 7.72%

The mathematically "optimal" threshold (0.2642) was deliberately not used -- its 3.19% false-accept rate is too high for a security-critical system. The deployed threshold of 0.40 forces near-zero false acceptance, accepting a higher false-rejection rate as the tradeoff: a false reject simply re-prompts the user within milliseconds, while a false accept is a real security failure.

3.2 Passive Liveness / Attack Detection

Re-measured on an expanded sample (n=76 attacks / n=75 genuine), superseding an earlier n=5 figure that was too small to trust.

At deployed threshold 0.90:

APCER (attacks wrongly accepted): 46.05% (35/76) -- worse than the earlier estimate
BPCER (genuine wrongly rejected): 0.00% (0/75)
ACER: 0.230

Per attack type:

Printed Photo	5.3% missed	- reliably caught
Screen Replay (Phone)	5.3% missed	- reliably caught
Screen Replay (Laptop)	73.7% missed	- REAL, CONFIRMED WEAKNESS
Frozen Frame	100% missed	- expected, handled by active liveness

A threshold sweep shows raising the deployed threshold to 0.95-0.96 is a free improvement in this sample (genuine users unaffected), but no threshold fully closes the laptop-replay gap without a real cost to genuine acceptance. This report does not change the deployed threshold -- that decision is left to the project owner with this evidence in hand.

3.3 Quality Assessment Profile Acceptance

Composite score combines 7 sub-checks (Contrast and Resolution added this pass).

Lenient (50% threshold):	100% acceptance (genuine captures)
Balanced (70%, default):	0% acceptance (small calibration sample)
Strict (85% threshold):	0% acceptance

4. Limitations and Scope Decisions - Stated Plainly

Physiological liveness (rPPG) - deferred. Built and calibrated offline (heartbeat detection from skin color) but not wired into the live pipeline; it needs a stable 10-15 second capture window that does not fit the current per-tick UI model.

Active liveness covers 2 challenge types (blink, head-turn). A direct test confirmed the expected consequence: a replayed video of a real blink is accepted (status: pass). Landmark-based active liveness cannot distinguish a live blink from a recorded one -- exactly why the system does not rely on it alone.

Occlusion detection is a visibility-based approximation, not a trained classifier. It can flag likely obstruction but not identify what is causing it.

Demographic bias - confirmed, unmitigated. Measured across 80 real images: quality pass rate 20.0% (female) vs. 42.5% (male); liveness pass rate 25.0% (senior) vs. 46.2% (middle-aged); a smaller but present skin-tone gap. No mitigation attempted.

Multi-angle enrollment simplified to front-only. The original three-angle flow was unreliable in practice, and duplicate-detection and verification already relied on the front template almost exclusively for accuracy.

A known, intermittent WebRTC connection hiccup exists -- the camera connection can silently drop for 20-30 seconds and self-recover. Root-caused as far as the evidence allows to a library-level characteristic, not an application bug.
Mitigation: keep continuous camera segments under about 30 seconds during live demonstrations.

5. Registration Quality Checks - Gap Closed This Pass

The brief specifies four registration image-quality checks: brightness, contrast, blur, and resolution. Brightness and blur existed from early in the project; contrast and resolution did not exist anywhere in the codebase, confirmed before writing any new code. Both have now been added, calibrated against real captured images, and wired into the composite quality score, the corrective guidance shown to the user, and the live camera guide overlay.

6. Verification - Fresh Automated Test Suite

Full suite run fresh for this report:

13 passed in 10.10s

Covering encryption round-trip integrity, consent enforcement, soft/hard delete behavior, embedding similarity math, template matching (best-of-angle and threshold rejection), and quality/liveness structure on genuine captures.

7. Learning Outcomes

Computer Vision: face detection and landmark geometry, solvePnP-based head pose recovery, and classical statistical quality proxies chosen deliberately over a heavier trained model.

Face Recognition: ArcFace embeddings, cosine similarity matching, and the distinction between a mathematically optimal threshold and one chosen for a security-first tradeoff.

Liveness Detection: a three-layer design motivated by the fact that single-method liveness is foolable -- demonstrated concretely by this project's own testing.

Biometric Authentication: consent gating, duplicate detection, encryption at rest, and GDPR/BIPA-oriented deletion, implemented as core requirements, not afterthoughts.

API Development: a FastAPI backend with key-based auth and rate limiting, kept cleanly separate from the core pipeline so both the API and UI share one source of truth.

Performance Evaluation: industry-standard metrics (APCER/BPCER/ACER, EER/ROC/AUC, HTER) rather than one invented accuracy figure, including the discipline of re-measuring a suspiciously small sample and reporting the result even when it came out worse than expected.

This report consolidates findings already documented in README.md, Evaluation_Report.md, and scope_decision_worksheet.md. No new analysis is introduced beyond what those source documents establish.